ECE250: Signals & Systems Quiz-10

Date: 19th November, 2025 **Total Marks:** 10 **Duration:** 15 Minutes

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Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
2. Institute Plagiarism policy are strictly applicable.

[CO2, CO4] [Q1]: Consider a continuous–time LTI system for which the input $x(t)$ and output $y(t)$ are related by the differential equation **[4 Marks]**

$$\frac{d^2y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t)$$

Let $X(s)$ and $Y(s)$ denote the Laplace transforms of $x(t)$ and $y(t)$, respectively, and let $H(s)$ denote the Laplace transform of $h(t)$, the system impulse response.

(a) Determine $H(s)$ as a ratio of two polynomials in s. Sketch the pole–zero pattern of $H(s)$.

(b) Determine the impulse response $h(t)$ for each of the following cases:

1. The system is stable. **[2 Marks]**
2. The system is causal. **[2 Marks]**
3. The system is neither stable nor causal. **[2 Marks]**